

SPECIFICATION — NODE 4 — PAGE 4

INSERTABLE INTERNAL GEOMETRY/COMPLIANCE DEVICE — NODE-SPECIFIC EXEMPLAR CLAIM-LIKE STATEMENTS (OPTIONAL, NON-LIMITING)

A. Overview

This page provides optional, non-limiting claim-like statements for Node 4 to support standalone protectability of the insertable internal geometry/compliance device. A provisional application does not require claims. These statements are illustrative only and do not limit the disclosure.

B. Exemplary Node 4 System Statements (Non-Limiting)

1. An insertable device comprising: a) a sealed body configured for intravaginal placement and, in certain embodiments, anal placement; b) an expandable stimulus element configured to apply controlled radial stimulus; c) a stimulus pressure sensor comprising a MEMS pressure transducer configured to measure a stimulus pressure; d) an optical endpoint sensor comprising an infrared time-of-flight (IR ToF) sensor configured to estimate an internal distance; and e) a circumferential contact sensing region configured to measure contact distribution and response; wherein a computing system computes at least one quality index and outputs privacy-preserving internal geometry and compliance metrics with confidence indicators. 2. The device of statement 1, wherein the expandable stimulus element comprises an annular inflatable cuff or microbladder ring. 3. The device of statement 1, further comprising a constraint band configured to shape or limit expansion. 4. The device of statement 1, wherein the device includes manual pneumatic actuation via a compressible handle element. 5. The device of statement 1, further comprising a micro-pump configured to deliver a controlled inflation profile. 6. The device of statement 1, wherein the circumferential contact sensing region comprises at least one of a piezoresistive/FSR ring, a capacitive pressure ring, or a piezoresistive film band. 7. The device of statement 1, further comprising a 6-axis IMU configured for motion gating and artifact rejection. 8. The device of statement 1, further comprising capacitive proximity/contact confirmation electrodes configured to confirm coupling. 9. The device of statement 1, further comprising a temperature sensor configured to provide thermal context and/or drift compensation. 10. The device of statement 1, wherein the computing system derives a compliance index from a relationship between stimulus pressure and circumferential contact response.

C. Exemplary Node 4 Method Statements (Non-Limiting)

11. A method comprising: a) confirming coupling of an insertable device using capacitive proximity/contact confirmation and/or circumferential contact sensing; b) applying controlled radial stimulus using an expandable element; c) measuring stimulus pressure using a MEMS pressure transducer; d) measuring contact distribution response using a circumferential contact sensing region; e) estimating internal distance using an optical endpoint sensor; f) computing at least one quality index including coupling integrity and motion stability; and g) outputting privacy-preserving internal distance and compliance metrics with confidence indicators. 12. The method of statement 11, further comprising generating a stimulus waveform comprising a ramp, a step, a hold, and a release. 13. The method of statement 11, further comprising rejecting segments based on IMU-derived motion gating. 14. The method of statement 11, further comprising computing a contact distribution signature comprising zone-based contact indices. 15. The method of statement 11, further comprising computing a compliance curve descriptor including hysteresis features based on inflation and deflation sequences.

D. Optional Integration Statements (Non-Limiting)

16. The system or method of any prior statement, further comprising time-aligning Node 4 outputs with derived outputs from at least one other node and computing an alignment score. 17. The system or method of any prior statement, further comprising using Node 4 compliance indices to condition cross-device correlation outputs under consent gating and quality thresholds.

E. Variations

Any statement herein may be modified to include additional sensors, fewer sensors, substituted equivalent modalities, different expandable elements, different actuation mechanisms, different housing geometries, different stimulus waveforms, and different processing distributions (on-node, on-device, cloud) while maintaining the functional intent of controlled stimulus-based internal geometry and compliance characterization with validity gating and confidence indicators.